Project 3: The Ising Model

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1 Introduction

The Ising model is one of the most important models in statistical physics. Originally created to model the behavior of ferromagnets, the Ising model has also been applied to many other systems involving local interactions, including other magnetic orders like antiferromagnets, binary alloys, and even fluids. This project explores the results of this model, using concepts such as the correlation function and block spin transformation.

2 The Metropolis Algorithm

One way to gain insight into a physical problem is to investigate it numerically. For the Ising model, it is not realistic to compute the probabilities of all possible configurations directly, since their number grows exponentially with system size. Instead, we use a Monte Carlo simulation, which generates representative configurations of the system using random sampling.

The basic idea is to evolve the system by randomly attempting to flip individual spins and deciding whether to accept or reject each change based on the resulting change in energy. At each step, a site on the lattice is chosen at random, and the change in energy associated with flipping that spin is

computed,

$$\Delta U = U_{\text{new}} - U_{\text{old}}.$$

If the change in energy is negative or zero, the spin flip is always accepted. If the change is positive, the spin is flipped with probability

$$P = e^{-\Delta U/T}.$$

This acceptance rule ensures that energetically favorable moves are always accepted, while energetically unfavorable moves are still possible, with a probability that decreases exponentially with the energy cost and increases with temperature. As the simulation is iterated many times, the system approaches thermal equilibrium and produces configurations distributed according to the Boltzmann distribution.

This procedure is known as the **Metropolis algorithm**, a version of the Monte Carlo method.

3 Problem 8.29

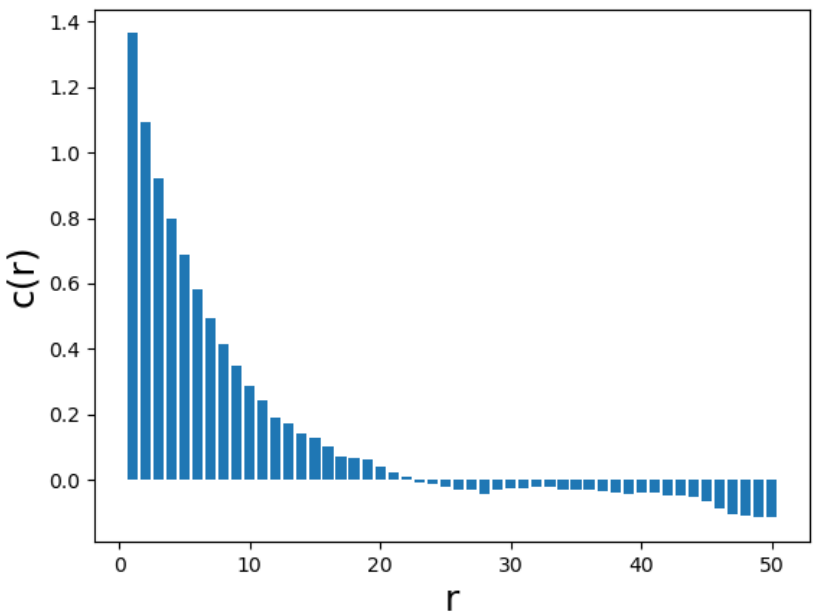
To quantify the clustering of alignments within an Ising magnet, we define a quantity called the **correlation function**, $c(r)$. Take any two dipoles, i and j, separated by a distance r, and compute the product of their states: $s_i s_j$. This product is 1 if the dipoles are parallel and -1 if the dipoles are antiparallel. Now average this quantity over *all* pairs that are separated by a fixed distance r, to obtain a measure of the tendency of dipoles to be "correlated" over this distance. Finally, to remove the effect of any overall magnetization of the system, subtract off the square of the average s. Written as an equation, then, the correlation function is

$$c(r) = \overline{s_i s_j} - \overline{s_i}^2$$

where it is understood that the first term averages over all pairs at the fixed distance r. Technically, the averages should also be taken over all possible states of the system, but don't do this yet.

(a) Add a routine to the `ising` program to compute the correlation function for the current state of the lattice, averaging over all pairs separated either vertically or horizontally (but not diagonally) by r units of distance, where r varies from 1 to half the lattice size. Have the program execute this routine periodically and plot the results as a bar graph.

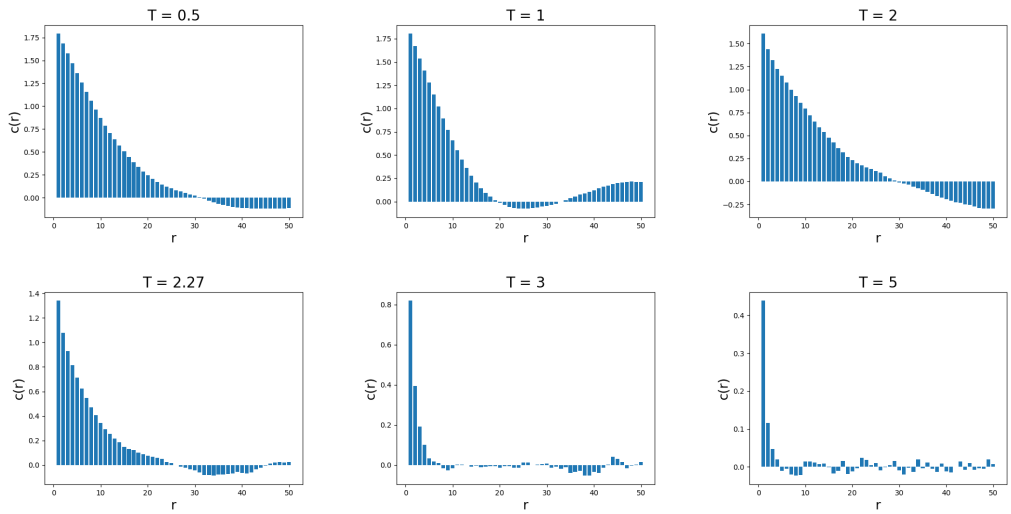
Answer: I ran the correlation for a lattice of size 100 after one simulation and created the following bar graph:



We can see the function decays exponentially as r increases and after around $r = 20$ it is roughly zero.

(b) Run this program at a variety of temperatures above, below, and near the critical point. Use a lattice size of at least 20, preferably larger (especially near the critical point). Describe the behavior of the correlation function at each temperature.

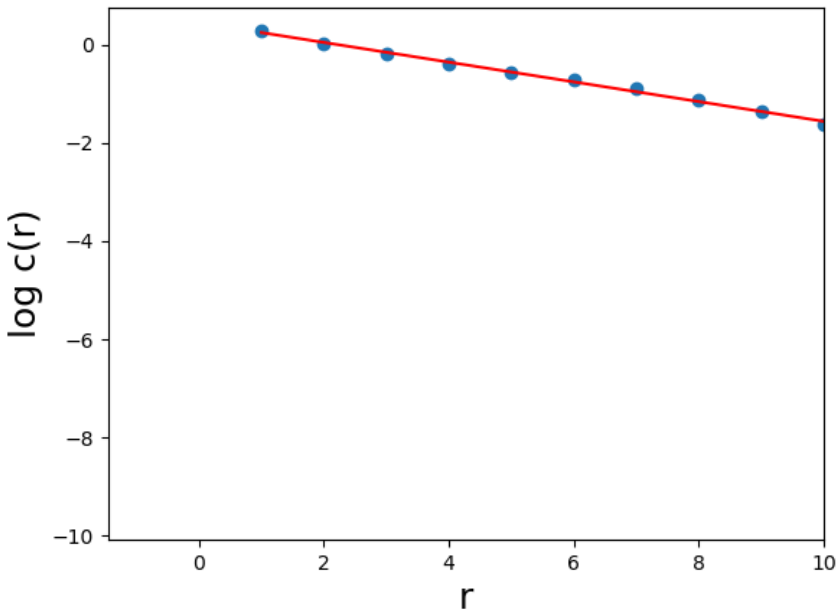
Answer: I created charts of the correlation for the following temperatures: 0.5, 1, 2, 2.27 (the critical temperature), 3, and 5:



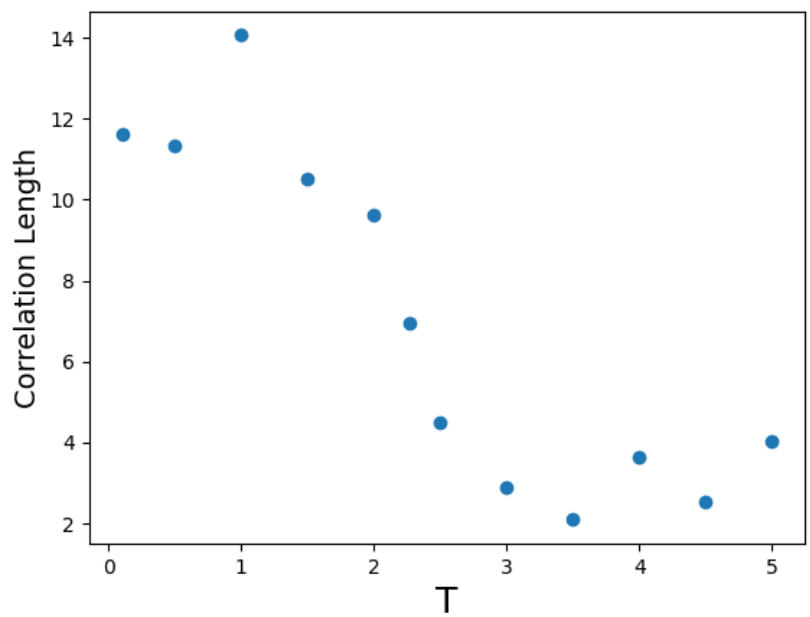
We can see that as temperature increases, the correlation function drops off much more rapidly with increasing r. At temperatures below T_c, there is significant correlation at large distances, which makes sense because this is the regime in which large scale structures appear. Above T_c, the correlation function drops to 0 rapidly because there are at most small structures, and the magnetization as a whole is largely uncorrelated.

(c) Now add code to compute the *average* correlation function over the duration of a run. (However, it's best to let the system "equilibrate" to a typical state before you begin accumulating averages.) The **correlation length** as defined as the distance over which the correlation function decreases by a factor of e. Estimate the correlation length at each temperature, and plot a graph of the correlation length vs. `T`.

Answer: By converting to a scatter plot and log scale, I can come up with a linear fit to estimate the correlation length, which should be the negative reciprocal of the slope:



I was able to plot correlation lengths for a series of temperatures, including the critical temperature:



It can be seen that the correlation length generally decreases with T. Past the critical temperature, the correlation length seems to bottom out between 2-4. This is telling us that below T_c, there is correlation across a relatively long distance, suggesting large structures. Above T_c, what correlation exists is only on short scales, implying small structures.

4 Problem 8.32

One of the most interesting features of the Ising model is the existence of a phase transition at a critical temperature T_c. Below the critical temperature the spins tend to align, producing a magnetized state, while above the critical temperature random fluctuations dominate and the spins become disordered. Near the critical temperature the system develops structures on many different scales, making the behavior difficult to analyze.

A powerful method for studying systems near critical points is the *renormal-*

ization group transform. The central idea behind the renormalization group is to examine how a physical system changes when we "zoom out" and ignore small scale details. In the Ising model this can be accomplished using a procedure called a *block spin transformation*.

4.1 Block Spin Transformations

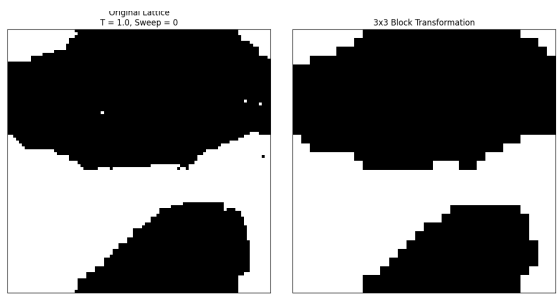


Figure 2: Example of a 3×3 block spin transformation applied to a 90×90 lattice at $T = 0$. Each 3×3 grid is replaced by a single site whose spin is the majority of the original block.

Instead of studying every microscopic spin individually, in a block spin transform, neighboring sites are grouped together into larger blocks and replaced by a single effective spin.

As example, we will divide the lattice into 3×3 blocks. Each block contains nine spins. The spins inside each block are replaced by a single new spin according to the *majority rule*:

$$S_{\text{new}} = \begin{cases} +1 & \text{if most spins point up,} \\ -1 & \text{if most spins point down.} \end{cases}$$

Since each block contains nine spins, ties are impossible.

Applying this transformation to the entire lattice reduces the size of the lattice by a factor of three in each direction. For example, a 90×90 lattice

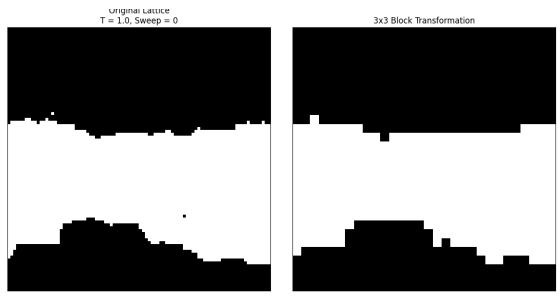
becomes a 30×30 lattice after one transformation as seen in Figure 2. Notice that after the block spin transformation, the transformed lattice resembles the structure of the original lattice, but with less precision.

As discussed earlier, the Ising model was simulated using the Metropolis algorithm. During each Monte Carlo sweep, a random spin was selected and flipped with probability

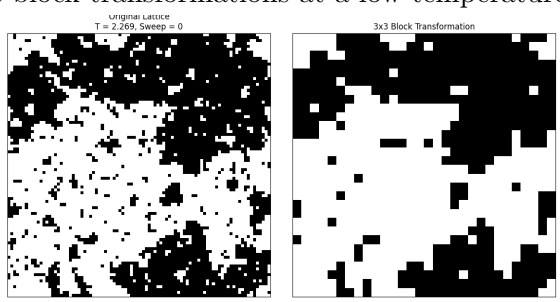
$$P = e^{-\Delta U/T},$$

whenever the energy increased by an amount $\Delta U > 0$. If the energy decreased, the flip was always accepted. After allowing the system to equilibrate, a block spin transformation was applied to the lattice. This was the process used to create Figure 2.

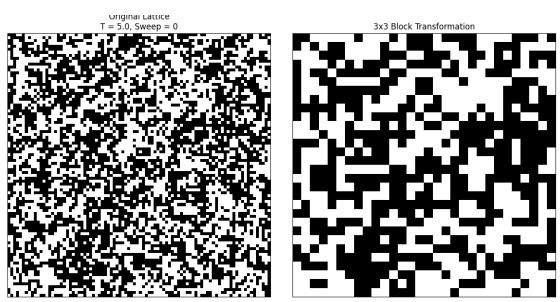
4.2 Equilibrating Systems



(a) Single block transformations at a low temperature ($T = 1$).



(b) Single block transformations near the critical temperature ($T \approx T_c$).



(c) Single block transformations at a high temperature ($T = 5$).

To study the effects of this transformation, simulations were performed on a 90×90 lattice at several temperatures above, below, and near the critical temperature.

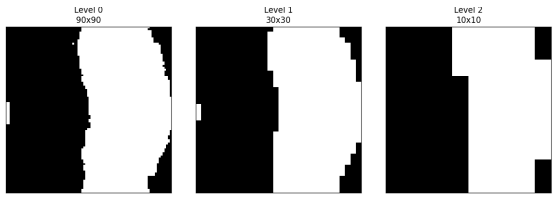
At temperatures well below the critical temperature, such as $T = 1.0$, the

lattice quickly evolved into a highly ordered state as seen in Figure 3a. Before the block spin transform was applied the system equilibrated into several large structures with similar spins, as predicted by the Ising model at low temperatures. After the block spin transformation was applied, the transformed lattice appeared even more ordered than the original lattice. Small isolated fluctuations disappeared during the averaging process. This behavior suggests that the transformed lattice corresponds to a lower effective temperature than the original system.

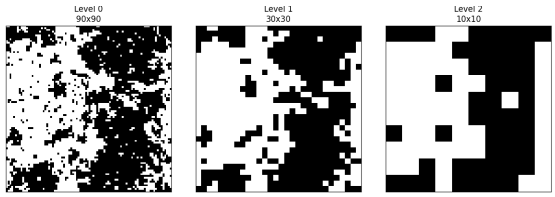
At temperatures well above the critical temperature, such as $T = 5.0$, the lattice appeared highly disordered as seen in Figure 3b. Before the transformation the system equilibrated into a disordered state with lots of small structures. The transformed lattice remained disordered after applying the transformation. In fact, the transformed lattice often appeared even more random because the majority rule averaged over already uncorrelated spins. This indicates that the effective temperature increases under the transformation when the original system is above the critical temperature.

Near the critical temperature, around $T_c \approx 2.269$, structures of many different scales formed. After the block spin transformation was applied, the transformed lattice looked very similar to the original lattice. This indicates that the transformation does not change the effective temperature for temperatures near the critical temperature.

4.3 Repeated Transformations



(a) Repeated block transformations at a low temperature ($T = 1$).



(b) Repeated block transformations near the critical temperature ($T \approx T_c$).

(c) Repeated block transformations at a high temperature ($T = 5$).

As seen earlier, applying the block spin transformation changes the effective temperature. Thus, if we apply the transformation several times to the same system, we should expect to see the effective temperatures converge to one of several fixed points.

If the initial temperature is less then the critical temperature then repeated transformations cause the system to become increasingly ordered. Structures of aligned spins grow larger while fluctuations become less important. Eventually, after enough transformations the effective temperature approaches $T = 0$. This can be seen in Figure 4a where after each block transformation the system looks like a system at a lower temperature, rapidly approaching zero effective temperature.

If the initial temperature is greater then the critical temperature then repeated transformations cause the system to become increasingly disordered. Correlations between neighboring spins disappear and the system approaches a completely random state. Eventually, after enough transformations the effective temperature approaches $T = \infty$. This can be seen in Figure 4c where after each block transformation the system looks like a system at a higher temperature, rapidly approaching infinite temperature.

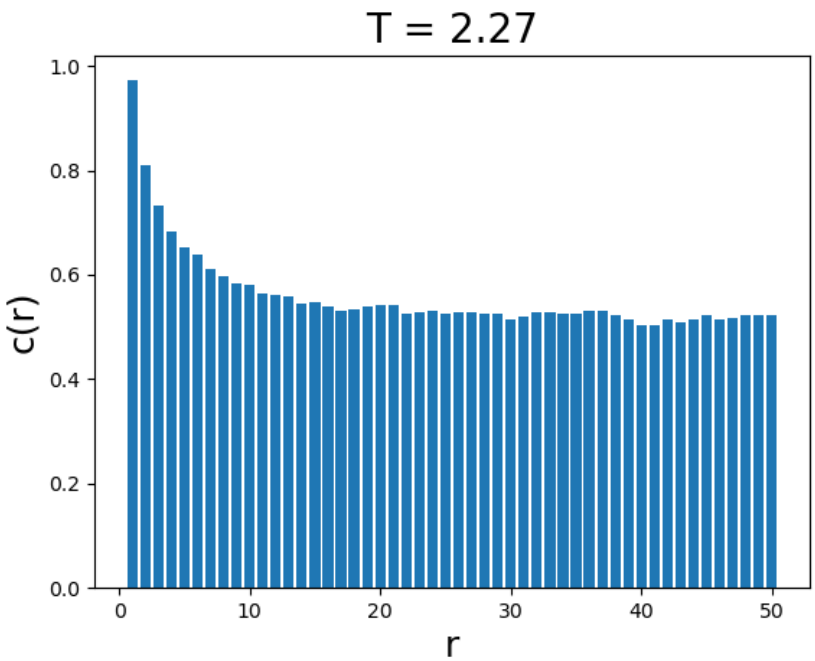
At the critical temperature, $T = T_c$, the system is approximately the same under repeated transformations. This occurs because near the critical temperature the block spin transformation does not qualitatively change the structures in the system. Because the system appears similar at different scales, the critical temperature is itself a fixed point of the renormalization group transformation. This can be seen in Figure 4b where after each block transformation the system looks to be at roughly the same temperature.

5 External Applied Field

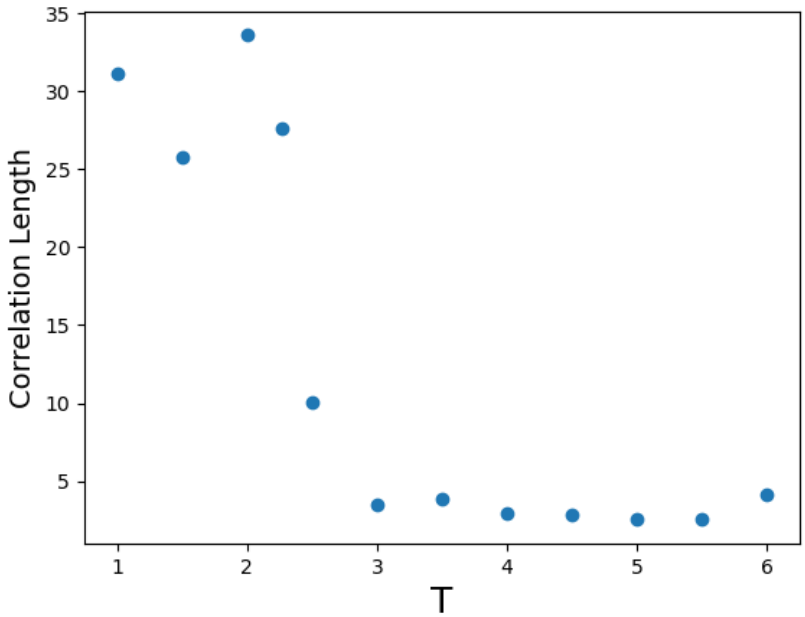
If an external magnetic field is applied to our lattice, it tacks on another term to our energy:

$$U = -\epsilon \sum_{i,j \ pairs} s_i s_j - B \sum_i s_i$$

In the Metropolis algorithm, this makes the state aligned with the magnetic field much more likely. This is reflected in the correlation function among other things. Consider the example of $T = 2.27$ now with a magnetic field $B = 0.05$:



In this case the lattice converged almost completely to a unified magnetization, and the correlation function also converges to just over 0.5. This means that over any length scale we have correlation, because almost the entire lattice is pointed in one direction. If we plot correlation lengths for many temperatures values with $B = 0.05$:



Interestingly, the critical temperature still appears to be at $T_c = 2.27$. The length in absolute terms increases in absolute terms below T_c but the dropoff occurs in the same place and above T_c the length again hovers between 2-4.

6 Conclusion

While the Ising model is not a perfect representation of the quantum mechanics in ferromagnets, it is impressive that the model predicts a phase transition at the critical temperature, just as real permanent magnets will break down above their Curie temperature. This T_c can be identified from the correlation function and from visual inspection of self-similarity. Manipulating the model through a block spin transformation or applying an external field can alter behavior above and below T_c, but the critical temperature is ultimately preserved.